

Cross-talker generalization: analysis and figure update

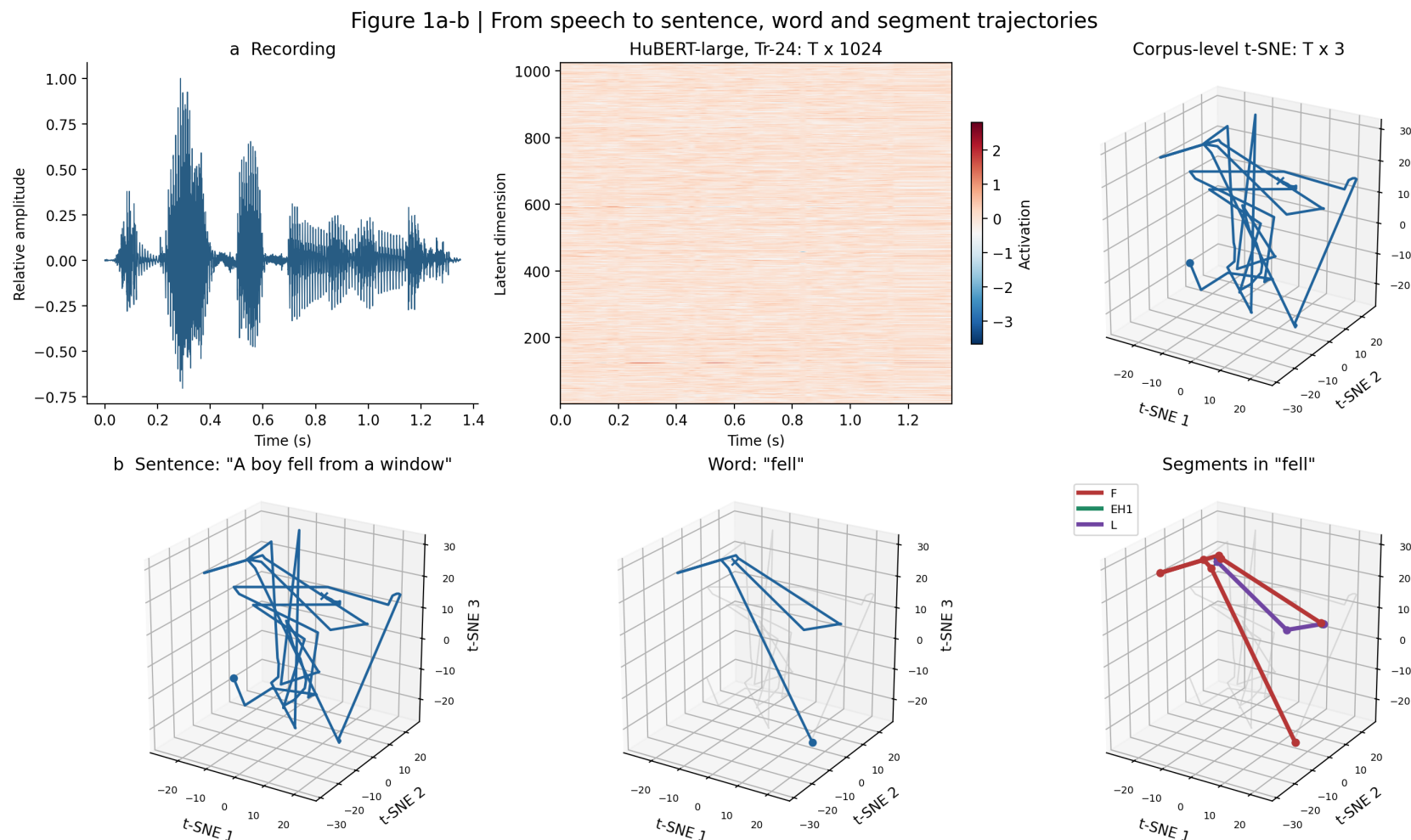
Zhengyang Jin | 6 September 2026

Following our September 1 meeting and your email, I have updated the figures and summarized where the analyses stand. I have organized this around the five points you raised.

1. **Optimization and reporting.** I have kept predictor-only likelihood and predictor z separate. For SBI, I show z across layers with three-fold intervals and a behavioral ceiling, alongside likelihood evaluation and both condition comparisons. I retain Tr-24 as the common-layer summary we discussed.
2. **Consistent figures.** I have brought together the layerwise z plots, conditional S-curves and model comparisons. I still need to recalculate a compatible ceiling for the revised HVE fits; the older notebook ceiling comes from a different fitting procedure.
3. **Figures 1 and 2.** I have added the Tr-24 speech trajectories, language coverage, a candidate PHOIBLE comparison, the 36-L2-talker matrix, and AN19/X21 control-condition curves. I have left 2b/2d open because AN19 has no existing phone-level labels.
4. **AN19 acoustic controls.** I found a coordinate-scaling mismatch between the component analysis and the full baseline. I need to rerun them with the same preprocessing before interpreting which components account for the strong AN19 result.
5. **Exposure understanding.** I would next like to test whether recognizing particular exposure sounds or words helps with related test words, while separating this from differences in listeners' overall ability.

Below I explain the figures and the remaining questions. I have included the new figures as PNG/SVG files with their underlying tables.

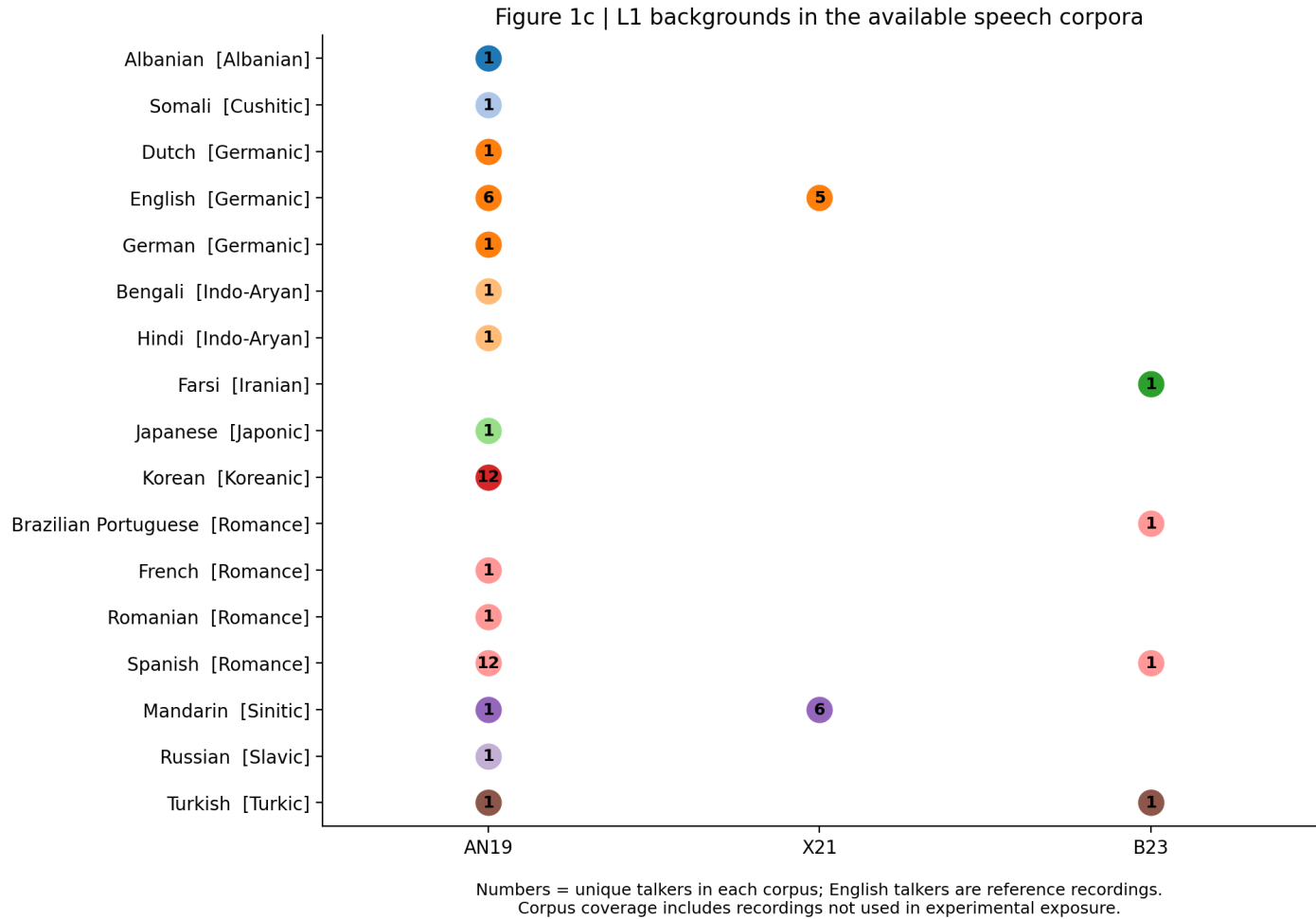
Figure 1a-b: from speech to latent trajectories



I used "A boy fell from a window," spoken by English-L1 male talker 055 in X21, to illustrate the transformation. The Tr-24 representation contains 67 frames with 1,024 dimensions before reduction and three afterwards. I used the existing t-SNE fit over all 660 corpus sentence segments (seed 42).

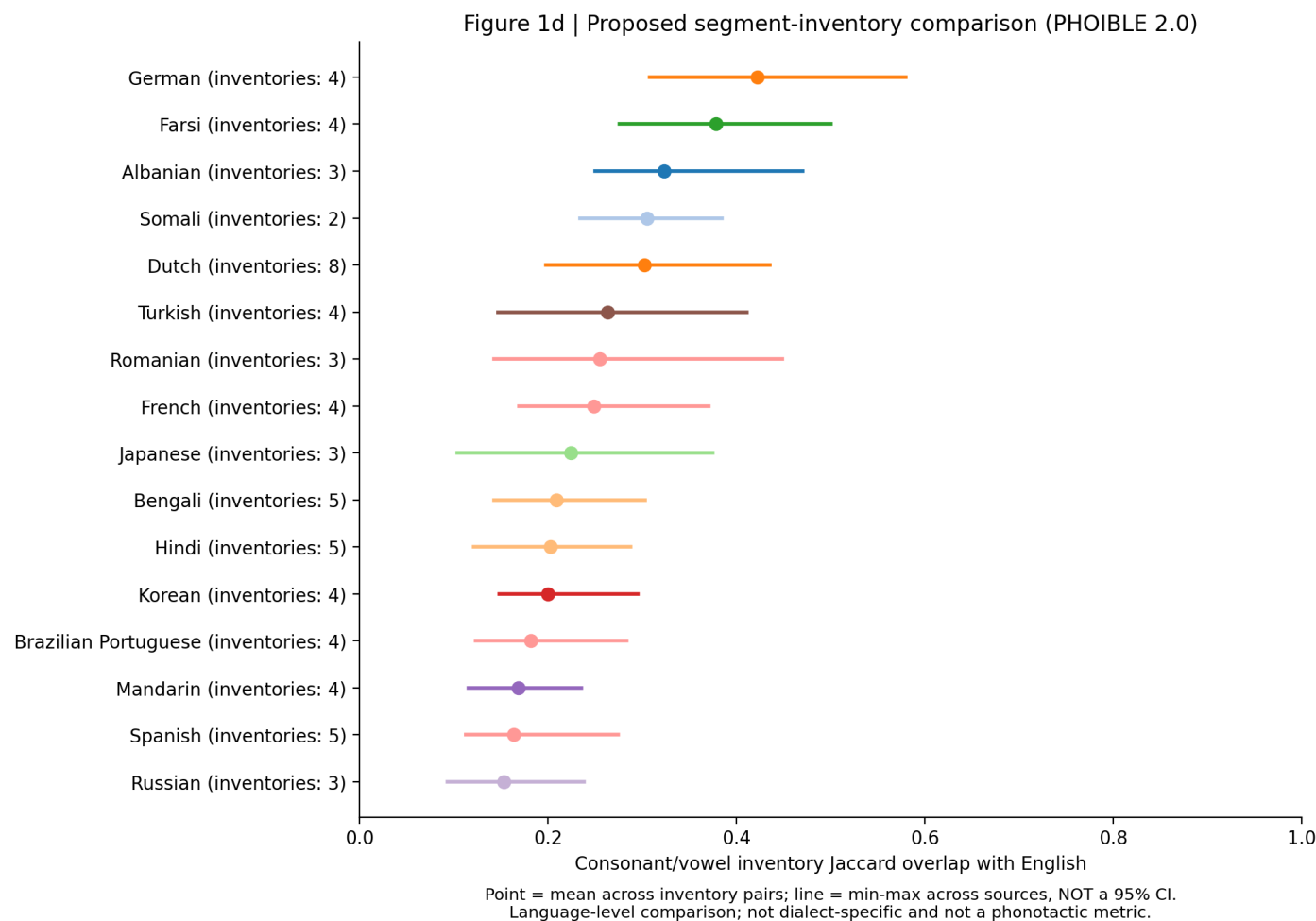
The lower row follows the sentence, the word "fell", and its segments F, EH1 and L using the available interval annotations. I kept all trajectories in the same coordinates and axis limits, without per-token normalization or smoothing. EH1 occupies one frame and appears as a point. Circles mark starts; crosses mark ends.

Figure 1c: L1 backgrounds across the datasets



I combined the L1 backgrounds across the three datasets, using colors for broad language branches. I changed the AN19 display labels "Indian" and "Chinese" to Hindi and Mandarin, following Table I of the original study. This gives 17 distinct L1 labels. Numbers indicate talkers in the available speech corpora, including English references; they do not indicate listener counts or exposure conditions.

Figure 1d: phonological inventory similarity to English

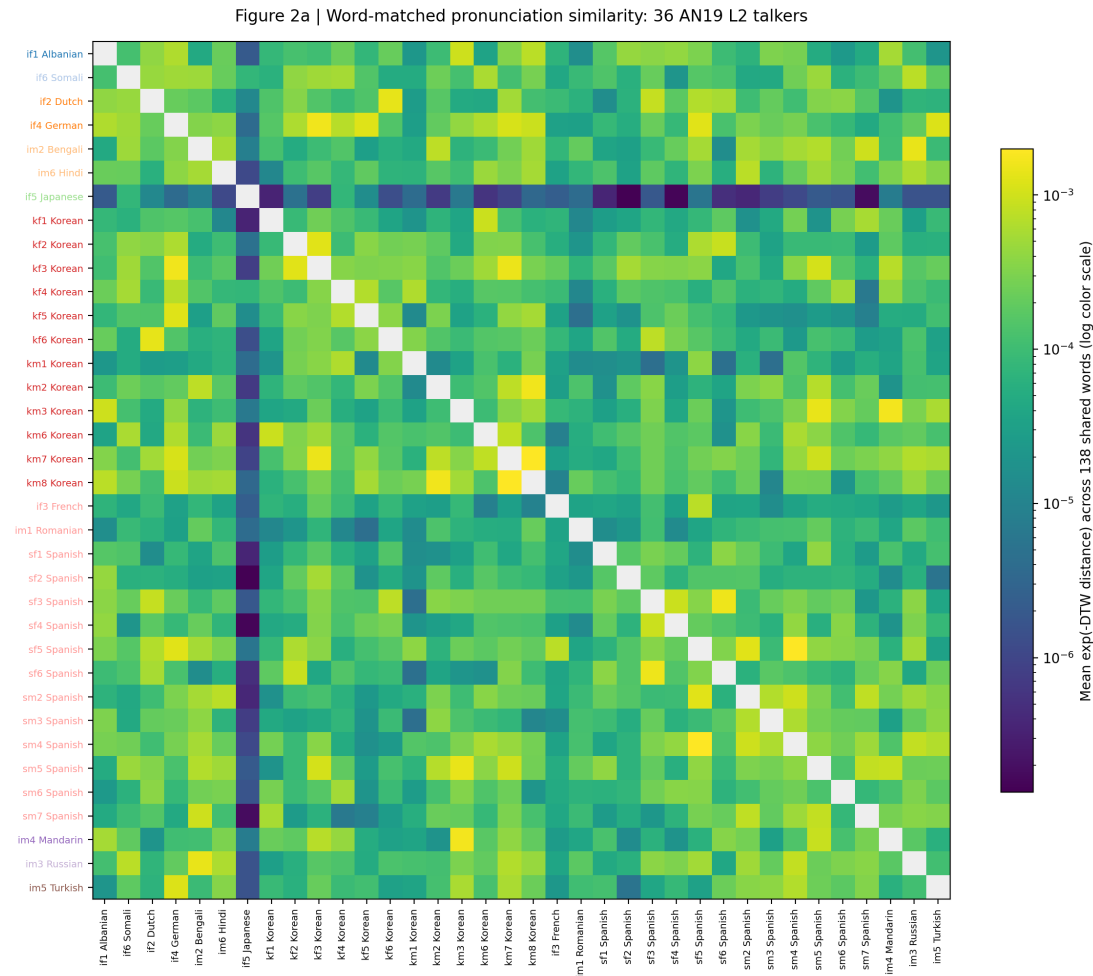


As a starting point, I calculated consonant/vowel inventory overlap with English using PHOIBLE 2.0. I used Jaccard similarity for each inventory pair and averaged over all available pairs. Points show these means; lines show the minimum and maximum across sources, not 95% confidence intervals.

This captures shared segment inventories, but not phonotactics or speaker-specific dialect differences, and it depends on transcription conventions. I would like your view on whether this narrower comparison is useful for 1d or whether we should add separate suprasegmental and phonotactic measures.

Source: Moran & McCloy (eds.), PHOIBLE 2.0, 2019, v2.0 release, CC BY-SA 3.0.

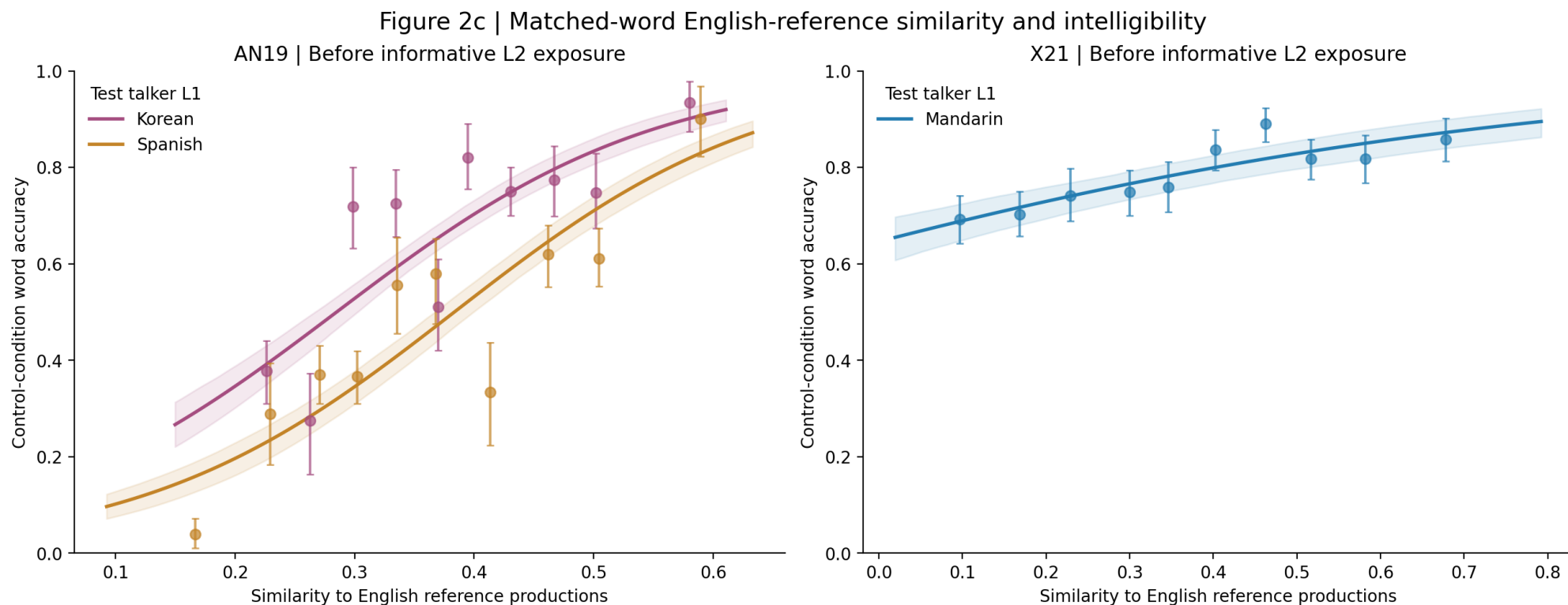
Figure 2a: pronunciation similarity among 36 AN19 L2 talkers



I selected the 36 L2 talkers from the original 42-talker matrix and grouped them by L1 and language branch. I use the six English talkers as references in Figure 2c instead.

Each cell averages $\exp(-\text{DTW distance})$ over the same 138 shared words, with equal weight per word. I retained the existing Tr-24 base-model distances, $\tau = 2$ and mean-sequence-length normalization. The color scale is logarithmic, and I have masked self-comparisons.

Figure 2c: English-reference similarity and control intelligibility

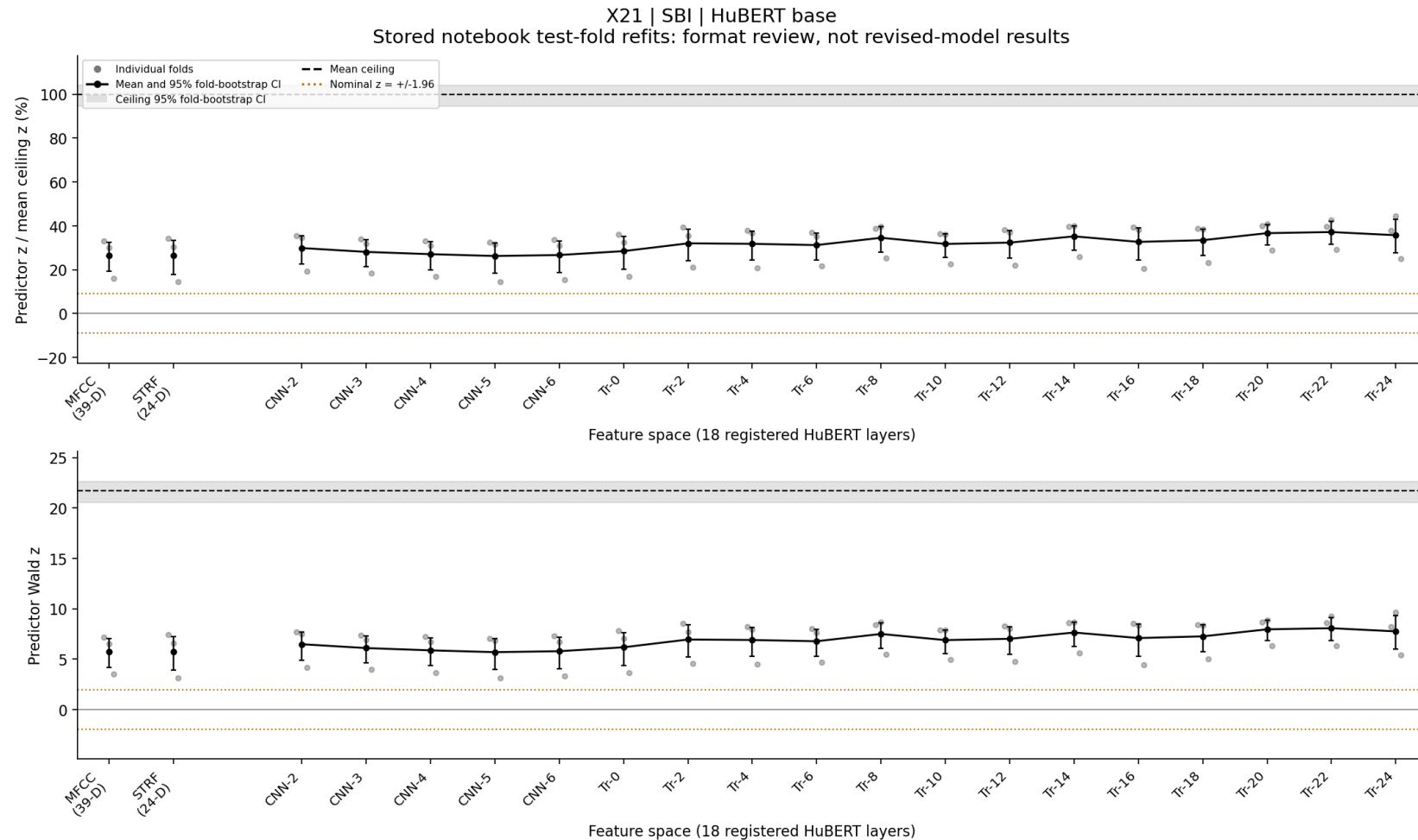


I compared each test word with the same word from six English talkers in AN19 or five in X21, averaging distances equally over reference talkers. For X21, I extracted word intervals from the sentence features. I converted distance to similarity using $\exp(-d/s)$, where s is the dataset median distance across unique matched targets, without using accuracy to choose the scale.

I see a positive association in all three accent groups. Points summarize ten within-accent quantile bins; curves are descriptive logistic fits. Both have 95% participant-bootstrap intervals (1,000 resamples), conditional on these items and talkers. These control-only curves are separate from the cross-validated GLMM tests.

I included 1,880/1,920 AN19 responses from 40 participants and all 4,117 X21 responses from 80 participants. I temporarily excluded 40 AN19 "wave" responses: the corresponding English hw74 recordings exist but are labeled "wade", while the L2 recordings say "wave". I need to resolve the spoken-word identity before treating these as same-word comparisons.

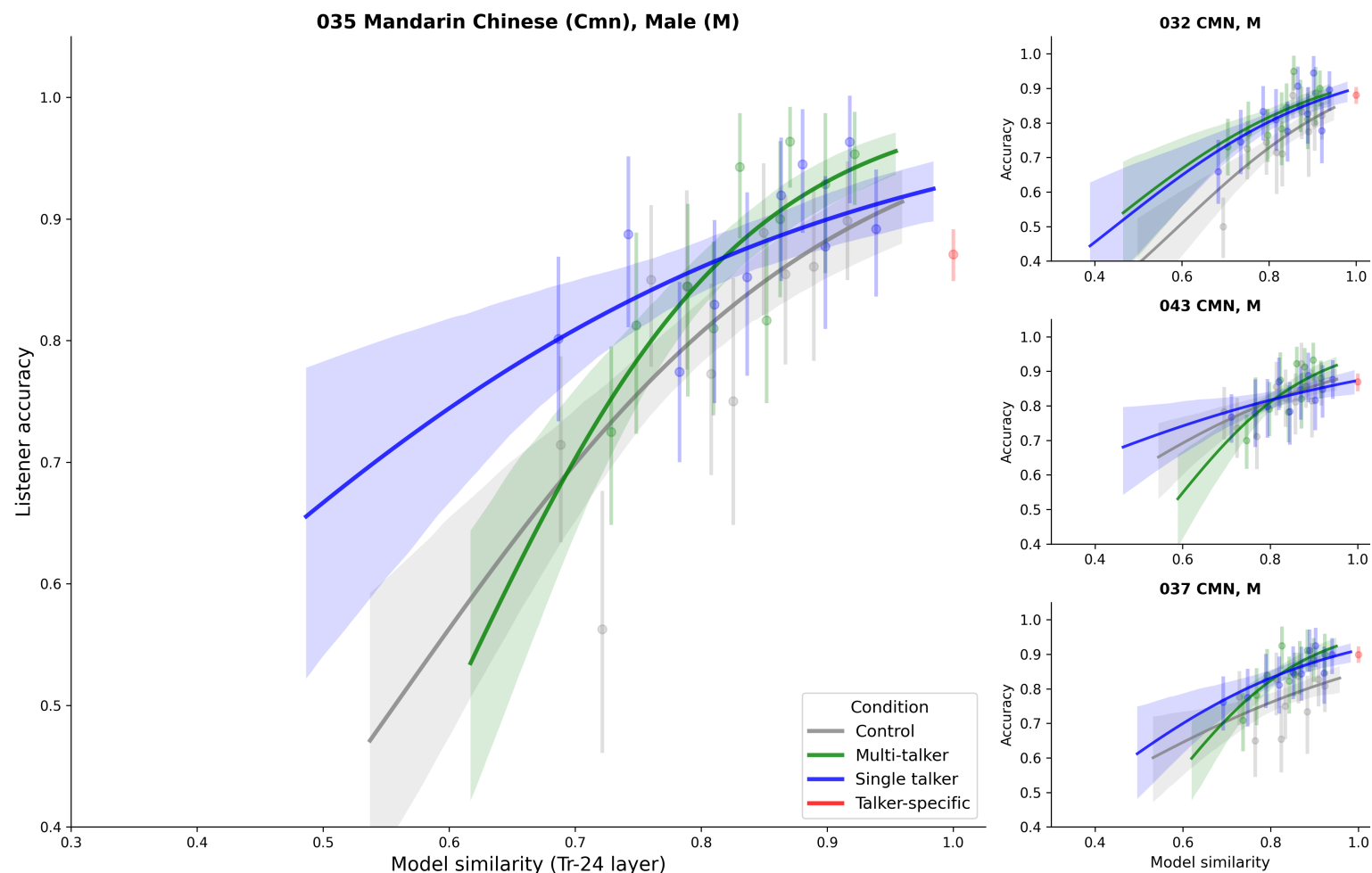
SBI: z across layers



I have put MFCC and STRF first, followed by the HuBERT layers. The three points show the folds, with their mean and 95% fold-bootstrap interval. In the upper panel, I set the mean ceiling to 100% and show its interval in gray; the lower panel gives raw z.

For this X21 example, I used the existing notebook results, where GLMMs were refitted within each test fold. These are not the revised-model fits or held-out prediction scores, and the normalized z is not a percentage of variance explained. The reference lines currently use nominal $z = \pm 1.96$. Before adding the Bonferroni line in the draft caption, I would like to agree which tests belong in the correction. The other SBI and HVE profiles are in the z gallery.

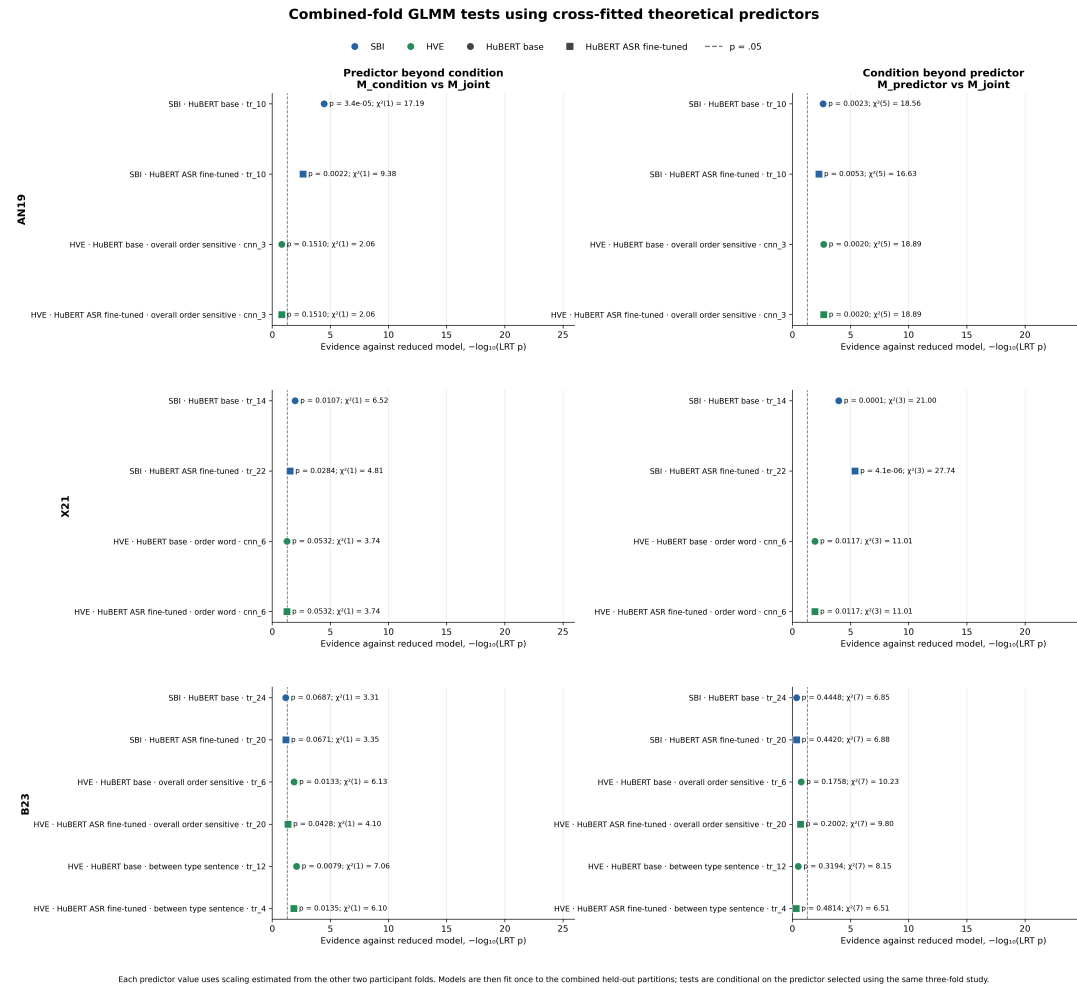
X21: similarity and accuracy by condition



I retained the condition-specific curves for each X21 test talker from my earlier notebook. Here, similarity is between exposure and test speech, unlike the English-reference similarity in Figure 2c.

The curves help show how the conditions differ at comparable similarity values. I use the joint-model comparison on the next page to test whether condition adds information beyond the theoretical predictor. The pooled-bin companion and source tables are in the August 21 package.

Does SBI/HVE add beyond condition, and vice versa?



On the left, I compare the joint model with the condition-only model to ask whether SBI/HVE adds information. On the right, I compare the joint model with the predictor-only model to ask whether condition adds information. I keep the theoretical predictor unchanged between these comparisons.

These are the September 1 likelihood-ratio tests fitted to the combined test folds using cross-fitted predictor values. Because I selected layers and methods using the same study, I treat these p-values as exploratory, alongside the held-out prediction results. This figure shows the selected layers, not the common Tr-24 summary.

What I still need to do

Analysis	What I have	Next step
Figure 2b	Word features, target transcripts, English references	Agree whether to add phone annotations and how to select phonemes independently of the results
Figure 2d	Written listener responses and the full caption	Use 2b deviations; align response errors and define minimal-pair neighbors with a pronunciation lexicon
Figure 1d	PHOIBLE inventory comparison	Discuss whether to extend beyond segment inventories
Updated z and ceiling	Revised HVE coefficients and older notebook z tables	Recalculate compatible ceilings and obtain current SBI coefficients for every registered layer
Objective comparison	Likelihood-based and z-based selection code	Compare B23 methods on the same participants, or keep the 97- and 168-participant sets separate
Acoustic components	Full baselines and initial component results	Rerun with matching coordinate scaling and response rows

I do not have phone-level labels for AN19, so I have put 2b/2d on hold. To pursue them, I would need to create and check the alignments first, then define the response-error and lexical-neighbor measures for 2d. I would like to discuss whether this additional annotation work should be a priority.

I also need to check the remaining ablations against the latest Overleaf version: ASR fine-tuning, architecture, full versus reduced representations, UMAP, PCA and similarity measures. I have kept the detailed source notes in `SOURCE_REVIEW.md` and the remaining tasks in `TODO`.